<u>Solution</u>: Work Sheet 7

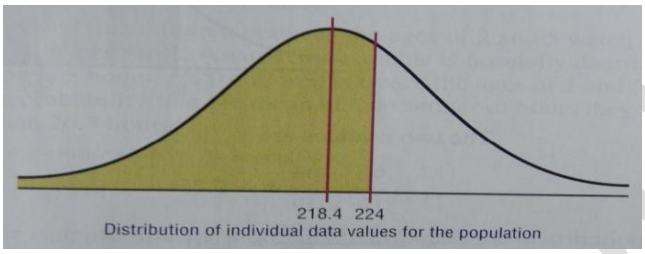
<u>Exercise 1</u>: The average number of pounds of meats that a person consumes per year is 218.4 pounds. Assume that the standard deviation is 25 pounds.

a) Find the probability that **<u>a person selected</u>** at random consumes less than 224 pounds per year.

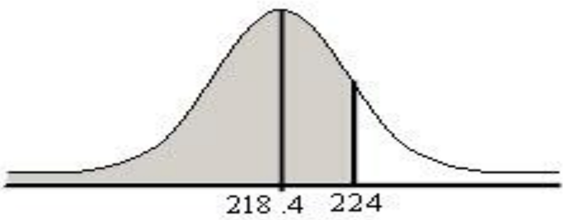
b) If a **<u>sample of 40</u>** persons is selected, find the probability that **<u>the mean</u>** of the sample will be less than 224 pounds per year.

<u>Solution:</u>

a. Since the question asked about an **<u>individual</u>**, so we use "**X**" and we need to calculate **P(X < 224)**, where X is a R.V. with mean $\mu = 218.4$ and $\sigma = 25$. Then we use: $Z = \frac{X-\mu}{\sigma}$



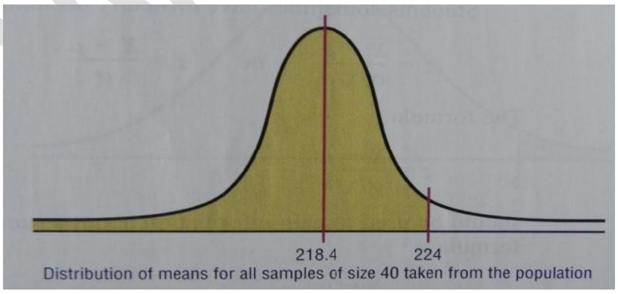
Distribution of individual data values for the population



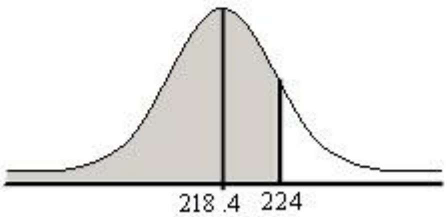
Then, $$P(X < 224) = P\left(Z < \frac{224-218.4}{25}\right) = P(Z < 0.22) = \Phi(0.22) = 0.5871.$$

b. Since the question asks about **<u>the mean</u>** of a sample with size "**n=40**", so we use "$\bar{X}$" and we need to calculate $P(\bar{X} < 224)$. Since "**n=40≥30**", so we can use the:

<u>Central Limit Theorem</u>: $Z = \frac{\bar{X}-\mu_{\bar{X}}}{\sigma_{\bar{X}}} = \frac{\bar{X}-\mu}{\left(\frac{\sigma}{\sqrt{n}}\right)}$, where $\mu_{\bar{X}} = \mu$ and $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$.



Distribution of means for all samples of size 40 taken from the population



Then, $$P(\bar{X} < 224) = P\left(Z < \frac{224-218.4}{\frac{25}{\sqrt{40}}}\right) = P(Z < 1.42) = \Phi(1.42) = 0.9222.$$

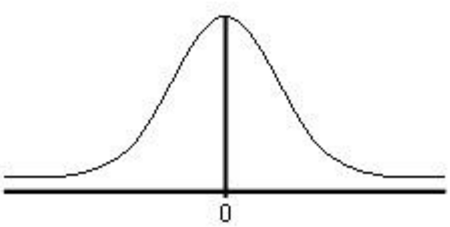
Exercise 2: The photo-resist thickness in semi-conductor manufacturing is normally distributed with a mean of 10 micrometers and a standard deviation of 1 micrometer. Suppose that the thicknesses of different wafers are independent.

- **a)** Find the probability that the **average** thickness of 10 wafers is **not** between 9 and 11 micrometers.
- **b)** Determine the **minimal number** of wafers that needs to be measured, so that the probability that the **average** thickness exceeds 11 micrometers is 0.01.
- **c)** What should be the **standard deviation** of thickness be equal to, so that the probability that the **average** of 10 wafers is either greater than 11 or smaller than 9 micrometers is 0.0001?

Solution: $\mu = 10$, and $\sigma = 1$

a. Let X denotes the thickness. Since **X is normally distributed**, then we can apply the central limit theorem (regardless the sample size which is n=10 in this case) for the sample mean.

We need to calculate: "**1 - P (9<$\bar{X}$<11)**", where $\mu_{\bar{X}} = \mu = 10$, and $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{1}{\sqrt{10}}$ = **0.316.**



- **First,** we calculate **P (9 < $\bar{X}$ < 11)** = P $(-\sqrt{10} < Z < \sqrt{10})$
  = P (-3.16 < Z < 3.16) = $\Phi(3.16) - \Phi(-3.16) = \mathbf{0.998422.}$
- **Then,** **1 – P (9 < $\bar{X}$ < 11) = 1 – 0.998422 = 0.001578.**

b. We need to calculate n =? (minimal number of wafers) such that:

P ($\bar{X}$ > 11) =0.01 and $\mu_{\bar{X}} = \mu = 10$, and $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{1}{\sqrt{n}}$

- $\Rightarrow$ **P (Z > $\sqrt{n}$)= 0.01 $\Rightarrow$ 1 - P (Z < $\sqrt{n}$)=0.01 $\Rightarrow$ P (Z < $\sqrt{n}$) = 0.99 $\Rightarrow$ $\Phi(\sqrt{n})$ = 0.99**

Z-table $\Rightarrow \sqrt{n} \approx 2.33 \Rightarrow n \cong 5.429$ Which is rounded up to **6**.

c. We need to calculate σ (**standard deviation of thickness**) such that

P (9 < $\bar{X}$ < 11) = 0.9999 and $\mu_{\bar{X}} = \mu = 10$, and $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{1}{\sqrt{10}}$; $n = 10$.

$$\Rightarrow P\left(-\frac{\sqrt{10}}{\sigma} < Z < \frac{\sqrt{10}}{\sigma}\right) = 0.9999 \Rightarrow P\left(Z < \frac{\sqrt{10}}{\sigma}\right) = 0.99995 \Rightarrow \Phi\left(\frac{\sqrt{10}}{\sigma}\right) = 0.99995$$

Z-table $\Rightarrow \frac{\sqrt{10}}{\sigma} = 3.89 \Rightarrow \sigma = \frac{\sqrt{10}}{3.89} = 0.8129.$

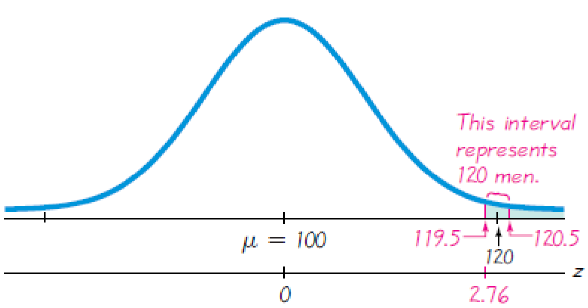
Exercise 3: An American Airlines Boeing aircraft has 200 seats. "When fully loaded with passengers, baggage, cargo, and fuel; the pilot must verify that the gross weight is below the maximum allowable limit, and the weight must be properly distributed so that the balance of the aircraft is within safe acceptable limits. Instead of weighing each passenger, their weights are estimated according to Federal Aviation Administration rules. In reality, we know that men have a mean weight of 172 lb, whereas women have a mean weight of 143 lb, so disproportionately more male passengers might result in unsafe overweight situation".

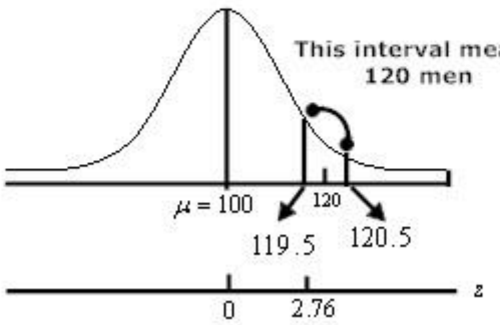
Assume that if there are at least 120 men in a roster of 200 passengers, the load must be somehow adjusted. Assuming that passengers are booked randomly, male passengers and female passengers are equally likely, and the aircraft is full of adults, find the probability that a Boeing with 200 passengers has at least 120 men.

Solution:

It is a **binomial distribution** "since we have only males or females" with **n=200, p=0.5, q=0.5** "because males and females are equally like".

We need to calculate **P (Y ≥ 120) =??**





i. **Check**: n*p=100≥**5** **&** n*q=100≥**5** **"both of them must be ≥ 5",** so we can use the normal distribution to approximate the required probability.

ii. $\boldsymbol{\mu = n * p = 100}$ & $\boldsymbol{\sigma^2 = n * p * q = 50}$, then $\boldsymbol{\sigma = 7.071067812}$.

iii. $\mathbf{P\ (Y \geq 120) \approx P\ (X > 119.5) = P\left(Z > \frac{119.5-100}{7.071}\right) = P\ (Z > 2.76)}$

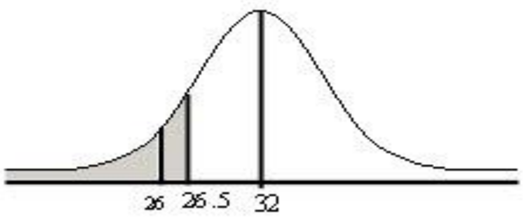
$\mathbf{= 1 - P\ (Z < 2.76) = 1 - \Phi(2.76) = 1 - 0.9971 = 0.0029}$

Exercise 4: If a baseball player's batting average is 0.32, find the probability that the player will get at most 26 hits in 100 times at bat.

Solution:

It is a **binomial distribution** with **n=100, p=0.32, q=0.68**.

We need to calculate **P (Y≤26) =??**



i. **Check**: n*p=32≥**5** **&** n*q=68≥**5** **"both of them must be ≥ 5",** so we can use the normal distribution to approximate the required probability.

ii. $\boldsymbol{\mu = n * p = 32}$ & $\boldsymbol{\sigma^2 = n * p * q = 21.76}$, then $\boldsymbol{\sigma = 4.6647}$.

iii. $\mathbf{P\ (Y \leq 26) \approx P\ (X < 26.5) = P\ (Z < -1.18) = \Phi(-1.18) = 0.1190.}$

Exercise 5: Suppose the diameter of metal rods is normally distributed with μ = ***112***mm and σ = ***2***mm. Suppose **500** rods are selected at random.

a) Find the probability that at least 50 rods have the diameter between 111 and 114 mm.
b) Use Poisson approximation: find the probability that exactly 2 have the diameters less than 107 mm.
c) Re-evaluate the probability in **a)** using normal approximation. Compare the results.

Solution:

a. Let:

- X denotes the diameter "continuous R.V." which is **normally distributed** with **μ = 112mm** and **σ = 2mm.**
- Y denotes the number of rods "discrete R.V." which is **binomially distributed**.

$$\boldsymbol{P(Y \le 50) = \sum_{k=50}^{500} \binom{500}{k} * p^k * (1-p)^{500-k}}$$

, where p=the probability that the diameter of one rod lies between 111 and 114

, i.e., **p = P (111 < X < 114) = P (-0.5 < Z < 1)**

$$= \boldsymbol{\Phi(1) - \Phi(-0.5) = 0.84134 - 0.30853 = 0.5328}.$$

$$\boldsymbol{P(Y \le 50) = \sum_{k=50}^{500} \binom{500}{k} * p^k * (1-p)^{500-k} = \sum_{k=50}^{500} \binom{500}{k} * (0.5328)^k * (0.4672)^{500-k}}$$

c. We need to re-evaluate the binomial probability **P (Y ≥ 50)?**

, where Y has n = 500 & p=0.5328 & q=0.4672

Now, since n*p = 266.4 ≥ 5 & n*q = 233.6 ≥ 5, then we can approximate that binomial probability using normal distribution for which:

$$\mathbf{\mu = n * p = 266.4\ \&\ \sigma^2 = n * p * q = 124.46, then\ \sigma = 11.156}$$

P (Y ≥ 50) ≈ P (X > 49.5) = P (Z > -19.44) = 1 – $\Phi(-19.44)$ = 1 – 0.001 ≈ 1

b. We need to calculate **P(Y=2)** by Poisson approximation, where

$$\boldsymbol{P(Y=2) = \frac{e^{-\lambda} * \lambda^2}{2!};\ \lambda = np\ \&\ n = 500\ \&\ p = P\ (X < 107)}$$

⇨ p= P (X < 107) = P (Z < -2.5) = 0.0062 ➔ $\boldsymbol{\lambda = (500)(0.0062)}$ **= 3.1**

⇨ $\boldsymbol{P(Y=2) = \frac{e^{-3.1} * (3.1)^2}{2!} = 0.2165}$.

✓ We can calculate **P (Y=2)** by Binomial distribution:

$\boldsymbol{P(Y=2) = \binom{500}{2} * p^2 * (1-p)^{498} = \binom{500}{2} * 0.0062^2 * 0.9938^{498}}$ = ***0.21663.***

✓ **This binomial probability CAN NOT be calculated or approximated by normal distribution because: n*p=3.1 < 5 although n*q > 5.**